

Week 1

```
rishi_alapati@BXTHub:~$ nano rishi.c
rishi_alapati@BXTHub:~$ gcc rishi.c
rishi_alapati@BXTHub:~$ ./a.out
Enter Linux command: ls
Parent PID: 13608
Child PID: 13609
a.out  pl.c  rishi.c
Child process completed
```

```
pranav_chowdary@Pranav: ~
GNU nano 8.7.1 q1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

[Read 27 lines]

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-C Copy

```
rishi_alapati@BXTHub:~$ uname -a
lscpu
lsblk
ps
top
Linux BXTHub 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13450HX
CPU family: 6
Model: 183
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
Stepping: 1
BogoMIPS: 5222.39
Flags: fpu vme de pse tsc mcr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp_lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 384 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 10 MiB (8 instances)
L3: 20 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
```

```
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swaps barriers and _user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSS-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
```

```
PID TTY TIME CMD
13544 pts/0 00:00:00 bash
14029 pts/0 00:00:00 ps
top - 09:59:00 up 24 min, 1 user, load average: 0.19, 0.10, 0.09
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7825.6 total, 7058.8 free, 600.0 used, 341.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7225.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24612	13376	9136	S	0.0	0.2	0:04.53	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3212	2140	2008	S	0.0	0.0	0:00.01	init
191	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.04	chronyd-starter
192	root	20	0	4472	3168	2924	S	0.0	0.0	0:00.00	cron
193	message+	20	0	9108	5844	4736	S	0.0	0.1	0:00.73	dbus-daemon
197	root	20	0	44092	29744	14612	S	0.0	0.4	0:00.16	networkd-dispat
201	root	20	0	18456	9544	8180	S	0.0	0.1	0:00.27	systemd-logind
234	root	20	0	5492	2700	2516	S	0.0	0.0	0:00.00	agetty
246	_chrony	20	0	24524	11912	7168	S	0.0	0.1	0:00.07	chronyd
253	root	20	0	123456	32560	17948	S	0.0	0.4	0:00.13	unattended-upgr
260	_chrony	20	0	12092	2456	1812	S	0.0	0.0	0:00.00	chronyd
622	rishi_a+	20	0	22472	11044	8600	S	0.0	0.1	0:00.26	systemd
624	rishi_a+	20	0	22908	4076	2076	S	0.0	0.1	0:00.00	(sd-pam)
740	root	20	0	8644	5556	4716	S	0.0	0.1	0:00.00	login
785	rishi_a+	20	0	6136	5392	3848	S	0.0	0.1	0:00.02	bash
913	polkitd	20	0	306444	8496	7604	S	0.0	0.1	0:00.14	polkitd

```
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5205	syslog	20	0	220548	4520	3664	S	0.0	0.1	0:00.04	rsyslogd
5330	root	19	-1	50528	14424	13196	S	0.0	0.2	0:00.09	systemd-journal
6442	root	20	0	33920	8168	6364	S	0.0	0.1	0:01.47	systemd-udev
7212	systemd+	20	0	22544	9924	7308	S	0.0	0.1	0:00.06	systemd-resolve
11421	rishi_a+	20	0	8548	4092	3740	S	0.0	0.1	0:00.00	dbus-daemon
13538	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader